

Stochastic Gradient Descent (SGD)

What is SGD?

Stochastic Gradient Descent is an optimization algorithm that updates the model's parameters using **one training data point at a time**.

Its goal is to **minimize the loss function**.

How it works

For each training row:

1 row

↓

Prediction

↓

Calculate error

↓

Calculate gradient

↓

Update weights

↓

Next row

↓

Repeat

The weight update is:

Where:

- $w \rightarrow$ model weight
- $\eta \rightarrow$ learning rate
- $L \rightarrow$ loss function
- Gradient $\rightarrow$ tells the direction of increasing loss

Example

Suppose:

Weight = 5

Gradient = 2

Learning Rate = 0.1

Then:

The weight changes from 5 → 4.8.

Why is SGD useful?

- Works well with **very large datasets**
- Uses less memory
- Updates very quickly
- Can start improving the model without processing the entire dataset first

Limitation

Because it uses only one row at a time, updates are **noisy**. The loss may go up and down instead of smoothly decreasing.